

4G LTE

Module 43

Duration: 18:28

Video: <https://www.youtube.com/watch?v=UGQrMVqHCMQ>

Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Dr Lal Chand, Department of Computer Engineering, Punjabi University, Patiala

Auto-generated YouTube transcript for personal study. Spoken wording belongs to the original lecture. Follow the video link above; this file is not an official CEC/SWAYAM publication.

0:00 [Music]

0:16 hello viewers I welcome you to the series of lectures on cyber security and information

0:23 security today we will discuss 4G LTE LTE stands for long-term

0:31 Evolution and it more applies for improving Wireless Broadband speeds to meet increasing

0:38 demand LTE is 4G Wireless Communications standard developed by the third

0:45 generation partnership project 3gpp 3gpp Engineers named the technology

0:53 long-term Evolution because it represents the next step in progression for

1:00 GSM a 2G standard to UMTS that is universal mobile telecommunications

1:07 service the 3G Technologies based upon GSM the goal of Lte was to increase the

1:16 capacity and speed of wireless data networks using new DSP that is digital

1:24 signal processing techniques and modulations it is designed to provide up

1:30 to 10x the speeds of 3D networks for mobile devices such as smartphones

1:36 tablets notebooks and wireless hotspots so now what is the goal a goal was the

1:44 redesign and simplification of the network architecture to an IP based system with

1:53 significantly reduced transfer latency compared to the 3G architecture evolution of 4G

2:00 LTE tracking the evolution of Wireless technology with the evolution we get

2:09 changes to the existing system as in

2:09 mobile network lot of evolution is occurring in 1973 the first call was made by

2:18 engineer

2:18 Martin Cooper from their Motorola Mobile phone after that in 1987 actor Michael

2:27 Douglas in the movie

2:27 Wall Street playing a Financial Titan talking over his mobile phone from the beach

2:37 the first smartphone Simon was

2:37 introduced to the world in 1992 as the first wireless device to the Imagine

functions Beyond making phone

2:45 calls Simon's Innovation was ahead of its time but it is not ready to handle Simon's
2:52 capabilities after the next two decades a series of mobile network technology
standards came to Market
3:00 the first generation of mobile Communications started with the advanced mobile
phone systems a
3:08 Ms it is a 1G from there we move to GSM
3:14 and CDMA 1 and then to UMTS and EV Doo
3:20 these are 3G Technologies the latest Technologies of 4G are LTE from the 3G
3:28 PP group and and 802.16m from i e in the case of
3:37 802.16m the candidate for 4G is also known as Wireless MN Advanced or ymax
3:46 2 LTE progresses through versions known as releases the latest release that
3:52 qualifies as being 4G is release 10 often called as LTE advanced
4:01 4 minutes, 1 second 4G LTE enables Advanced Technologies introduced from the
last three years in transportation healthc care small
4:09 business Enterprise and education Solutions like intute go payment enable
consumers and businesses
4:19 to experience the power of 4G LTE through go payments small businesses
4:25 such as Street vendors farmer markets and food trucks can attach a small card
reader to a smartphone or tablet making
4:34 payment transactions quick and easy for the vendor and customer with the last 40
Years of mobile technology in the rear view
4:43 mirror and the large scale deployment of V LTE and LTE Advanced on the horizon
4:51 one can only imagine what the next 40 years will hold LTE technical
4:57 objectives user through port in megahertz down link Uplink down link
5:06 capacity Uplink capacity latency that is transition time less than 5 millisecond in
ideal
5:14 conditions that is user plan 100 millisecond control plane fast connection setup
simplified
5:23 architecture simpler e u t r n architecture no RNC C no CS domain no
5:33 DCH Mobility optimized for low speed but supporting 120 km/ hour now what is the
5:42 difference between 4G and LTE 4G LTE has several competing 4G
5:49 standards with ultra mobile broadband UMB and yaax i e
5:57 802.16 4G Technologies are designed to provide IP based voice data and
6:04 multimedia streaming at speeds of at least megabit per second and up to as
6:10 fast as 1 gbit per second the leading cell providers has started to deploy 4G
6:17 Technologies with verizone and AT and T launching 4G LTE networks and Sprint
6:25 utilizing its new 4G ymax network in terms of mobile devices most of the
6:32 newer Android based smartphones are 4G LTE capable the iPhone 5 and iPad 3 are

6:41 expected to have built-in 4G LTE capabilities when released in the second half of
6:49 2012 speed the difference between slow 3D networks and new 4G or LTE networks
6:57 is faster most of the 4G and true 4G networks have upload and download speeds
7:04 that are almost identical but now LT is the fastest connection available for wireless
7:11 networks resources must for 4G LTE technology for creating 4G connectivity
7:20 it requires two components first Network that can support the necessary speeds
7:27 second device that is able able to connect to that Network and download
information at a high enough speed but
7:36 just because a phone has 4G LTE connectivity inside doesn't mean you can get the
speeds you want in the same way
7:44 that buying a car that can drive 200 m per hour doesn't mean you can go fast on
7:51 55 m per our freeway is LT is same as 4G
7:57 or not no LT is not same as 4G there is 4G and then there is 4G
8:06 LTE 4G stands for fourth generation which comes next to the previous 3G networks
it is much faster and with a
8:16 greater capacity 4G LT stands for fourth generation long-term Evolution which is
8:24 the purest form of 4G technology as if a cellular provider describes a 4G network
without
8:32 mentioning LTE they are probably talking about a high speed packet exess
8:39 Network the HSPA network is a faster version of 3G gsm network while not as
8:47 fast as LTE network it is still faster than a 3G network advantages of
8:55 Lte provides low latency High Network throughput increased data transfer
9:03 speed more cost Effectiveness improvements over 3G
9:10 network devices which are covered under LTE or mobile phones laptops cameras
9:19 camcorders assured interoperability with older Wireless Technologies such as GSM
9:27 wcdma HSP a CDMA TDS CDMA LT will ensure
9:34 the success of mobile internet difference between different Generations 4G stands
for fourth
9:42 generation of Cellular Communications 10 times faster than current 3G so one
don't need Broadband in fact many things
9:51 could be done through smartphones the main reason 4G is faster than 3G is
because of orthogon frequency
10:00 division multiplexing that is ofdm ofdm is a technique for squeezing
10:08 more data onto the same amount of radio frequency it also reduces latency and
10:15 interference 4G networks use different frequencies to transmit data than 3G so
10:22 you need a handset which has a modem that supports these new frequencies LTE is
a 4G Wireless
10:31 Broadband technology developed by the third generation partnership project 3G

10:37 PP LT provides significantly increased Peak data rates with the potential for
10:43 100 megabits per second Downstream and 30 megabits per second Upstream
reduced
10:51 latency scalable bandwidth capacity and backwards compatibility with existing
10:57 GSM and um s technology VTE it enables Wireless carriers to use
11:06 IP based 4G LT data networks for voice calls like WhatsApp call video call
11:13 conferencing Etc 5G fifth generation of cellular
11:20 communication 5G networks are aimed at meeting the requirements for mobile
Communications Beyond 2020
11:29 the speed of 5G is much faster than 4G 20 GB per second fast enough to
11:36 download HD movies in a span of seconds so all of the above are expected to be
driver for Trends such as the
11:44 internet of things and initiatives such as smart cities and digital India and
revolutionize India a speedy way to
11:53 reach our 2020 vision of apj clam disadvantages of 4G
12:00 LTE it has advanced data rates for customers customers are required to purchase a
new equipment to maintain the
12:09 4G the 4G is only presently accessible in convinced cities needs complex
12:17 Hardware high definition voice the most noticeable benefit consumers will
experience with
12:25 volte is improved voice quality traditional voice networks transmit voice calls using
a 8 kilobits per
12:34 second codec Varone says it's using a 13 KB PS codec that also uses more modern
12:44 compression methods Rich communication services or RCS
12:51 VTE also will enable wireless operators to deliver a new set of Standards Based
12:58 Services refer to as Rich Communications services or RCS these Services include
things like
13:06 video calling file transferring realtime language translation video voicemail and
13:14 instant messaging many of these applications are already delivered over cellular
data service using the third
13:21 party apps for instance iPhone users are very familiar with FaceTime for video
13:28 calling millions of people use Skype for video calls as well as messenger services
and Google offers it Google
13:37 translate service the benefit of VTE is that it lets you launch these
13:44 Services directly from the phone's native dialer faster call setup times
13:51 connecting calls using volte will be much faster than it is today using the older
circuit switch technology on a 2G
13:59 wireless network improved battery life over other vo IP apps a report from signals
research

14:08 group suggests that VTE services offer much improved battery life compared with
14:15 using an over the-top voice over IP service such as Skype VTE delivered a
consistently higher call
14:25 Quality than circuit switched voice and over the top applications in addition
14:33 VTE really shined when it came to its Network requirements consuming far less
14:40 than a Skypey voice call which translates into a meaningfully longer smartphone
battery
14:48 life integration of VTE with voice over Wi-Fi
14:54 service another benefit for consumers is better integration between a wireless
operators cellular voice service and a
15:03 Wi-Fi calling service this is particularly important for operators which are starting
to use
15:10 Wi-Fi to handle some voice calls as well as other data traffic improving the
transition between
15:18 the cellular voice call which uses licensed Wireless spectrum and a Wi-Fi call which
uses unlicensed spectrum is
15:28 important because it could let these operators get better indoor coverage in areas
where their licensed Wireless
15:37 spectrum is limited that means improved coverage for consumers because the call
15:43 setup necessary for voice over Wi-Fi is the same as the setup for
15:50 VTE once volte is fully deployed it will be easier and more seamless to offer voice
over Wi-Fi in
15:59 areas where licensed spectrum is either weak or unavailable true device
16:06 interoperability once carriers make VT Services interoperable among different
carrier networks there won't be any
16:16 technology barriers among most of the world's Wireless operators the CDMA GSM
divide that separated the four major
16:25 Wireless operators in the US will no longer be an issue what this could mean for
the average consumer is that you may be able
16:34 to buy one device that could truly operate on any wireless network all the major
carriers would be delivering data
16:42 services and voice services using the same 4G LT technology and if your
smartphone supported enough LT Wireless
16:52 bands you would be able to take that device to any carrier and get both voice and
data service service now security of
17:01 17 minutes, 1 second 4G LTE LTE security it is a term that
17:08 provides a powerful defense mechanism versus possible threats from the internet
imposed by multiple types of
17:16 attacks confidentiality means making sure that the right people can in fact get the

17:24 right data integrity means the message sent from party a to the party B did not
17:32 change these are the essential points to highlight the security issues of LTE
technology for the Next Generation in
17:40 Mobile Communication in order to provide more powerful security and that will be
done through achieving the sub
17:48 objectives to discover the main factors that used to compare the LTE cryptographic
17:56 algorithms to find out these trends and weaknesses of each LTE cryptographic
18:02 algorithm to evaluate which algorithm provide higher security than others to
18:09 propose an algorithm from the existing LTE cryptographic algorithms and that may
enhance the security of the existing
18:18 LD cryptographic algorithms so that's all for 4G LTE thank you viewers for
watching this
18:26 program